

MultiMinds at MAHED 2025: Multimodal and Multitask Approaches for Detecting Emotional, Hate, and Offensive Speech in Arabic Content

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Abstract

This paper describes the MultiMinds team’s participation in the MAHED 2025 shared task at ArabicNLP 2025, which targets the detection of hate speech, hope speech, and emotional expression in Arabic content. We addressed two subtasks. For the text-based subtask (Task 2), we experimented with multiple models, including Support Vector Machines with TF-IDF and AraBERT embeddings, XGBoost with fused AraBERT and XLM-RoBERTa embeddings optimized via Optuna, and a fine-tuned AraBERT model and GPT-5 (gpt-oss-20b). The fine-tuned AraBERT achieved the best performance with an F1 score of 0.68. For the multimodal subtask (Task 3), we proposed an architecture combining DistilBERT for text representation with a lightweight ELU-Net enhanced by a cross-attention mechanism, reaching 75% accuracy. Major challenges included dataset imbalance and noisy text, which we mitigated through preprocessing, class-weighted optimization, and feature fusion. Our results demonstrate the benefits of combining multiple embedding layers for text classification and leveraging lightweight multimodal architectures for robust hate speech detection in Arabic.

1 Introduction

Online media has become an important avenue for the consumption and distribution of information, and many people now rely on it as their primary source of news (Perrin, 2015). These have enabled individuals to share their views effortlessly through images and texts (multimodal and/or unimodal), reaching a broad and diverse audience (Fortuna and Nunes, 2018). With the rapid increase in media posts, manual detection of emotion, hate, and offensive (EHO) content becomes impractical. Consequently, there is a growing interest in developing automated methods for EHO detection.

MAHED 2025 (Zaghoulani et al., 2025) is a shared task at ArabicNLP 2025 Co-located with

EMNLP 2025, focusing on the detection of hate speech, hope speech, and emotional expression in Arabic content. Participants may choose to participate in one or more of the following three subtasks: (i) Text-based Hate and Hope Speech Classification, (ii) Emotion, Offensive, and Hate Detection (Multitask), and (iii) Multimodal Hateful Meme Detection. We, MultiMinds, participated in MAHED 2025, with particular interest in tasks (ii), (iii) and ranked 10th and 7th, respectively.

For Task (ii), three methods were tested for Arabic emotion, offensive, and hate-speech classification: Support Vector Machines (SVM) as a Baseline model with TF-IDF (best macro F1: 0.517); XGBoost with TF-IDF, AraBERT embeddings, and fused AraBERT and XLM-RoBERTa embeddings, which were optimized via Optuna (best F1: 0.57); and a deep learning approach fine-tuning AraBERT, which achieved the highest performance score. As the dataset was imbalanced and contained unnecessary information, the key challenge was to extract the correct information from the text. In our experiment for Task (iii), we used 1D-CNN model (Singh et al., 2021) as the Baseline model by extracting image and caption features by CLIP processor. Our enhanced ELU-Net architecture got the best results by incorporating a cross-attention mechanism to combine visual and textual features generated from the DistilBert (Sanh et al., 2019) tokenizer. Full Implementation here - [Github](#). The main challenge of this task was that the classes were not equally distributed. Our key findings were as follows.

- Fusing multiple embedding layers from different textual models improves data representation.
- Using class weights enhances results.
- First-time use of a lightweight multimodal model to classify hateful and non-hateful memes.

2 Background

2.1 Emotion Detection

In recent years, research into developing state-of-the-art models for Arabic natural language processing tasks has gained momentum. [Alswaidan and Menai \(2020\)](#) proposed three models for emotion recognition in Arabic text. [Abdullah et al. \(2018\)](#) described their system - SEDAT, and showed substantial improvements in Spearman correlation scores over the baseline models. [Alsmearat et al. \(2015\)](#) explored the Gender Identification (GI) problem for Arabic text as a supervised learning problem and compared the Bag-Of-Words (BOW) approach with computing features related to sentiments and emotions. [Biswas and Zaghouani \(2025b\)](#) introduces a bilingual dataset comprising 23,456 entries for Arabic and 10,036 entries for English, annotated for emotions and hate speech, addressing the scarcity of multi-emotion (Emotion and hate) datasets. [Al-Henaki et al. \(2025\)](#) introduced MultiProSE, an open-source extension of the existing Arabic propaganda dataset, ArPro, with the addition of sentiment and emotion annotations for each text.

2.2 Offensive And Hate Speech Detection

While social media promotes free expression, it also fosters environments where hate speech spreads, making its detection a key research priority. [Alsafari et al. \(2020\)](#) built a reliable Arabic textual corpus by crawling data from Twitter. [Mubarak et al. \(2023\)](#) introduced a generic, language-independent method to collect a large percentage of offensive and hate tweets. [Aldjanabi et al. \(2021\)](#) developed a classification system for determining offensive and hate speech using a pre-trained Arabic language model. [Biswas and Zaghouani \(2025a\)](#) introduces multilabel hate speech dataset with offensive content in the Arabic language. [Zaghouani et al. \(2024\)](#) analyzes 70,000 Arabic tweets, from which 15,965 tweets were selected and annotated, to identify hate speech patterns and train classification models.

2.3 MultiModal Hate Speech Detection

The usage of social media has enabled individuals to disseminate hateful messages through the use of memes. [Chhabra and Vishwakarma \(2023\)](#) highlighted handcrafted feature-based and deep learning-based algorithms by considering multimodal and multilingual inputs. [Alam et al. \(2024a\)](#)

explored the intersection between propaganda and hate in memes using a multi-agent LLM-based approach. [El-Sayed and Nasr \(2024\)](#) described an approach to hateful meme classification for the Multimodal Hate Speech Shared Task at CASE 2024. [Arya et al. \(2024\)](#) introduced a novel approach by leveraging the CLIP model, fine-tuned through the incorporation of prompt engineering. [Alam et al. \(2024b\)](#) focused on developing an Arabic memes dataset with manual annotations of propagandistic content. [AIDahoul and Zaki \(2025\)](#) explores the potential of large language models to effectively identify hate speech, offensive language, and emotional expressions. [Kmainasi et al. \(2025\)](#) introduced MemeIntel, an explanation-enhanced dataset for propaganda memes in Arabic and hateful memes in English. However, multimodal hate speech detection lacks the use of lightweight architectures.

3 System Overview

Before tackling Task 2, we observed that the dataset ([Zaghouani et al., 2024](#)), ([Biswas and Zaghouani, 2025b](#)), ([Biswas and Zaghouani, 2025a](#)) was both imbalanced and noisy. To address the noise, we performed text cleaning and preprocessing, converting the text into TF-IDF features and tokenizing it using the AraBERT tokenizer. We then fused the embedding layers of XLM-RoBERTa ([Conneau et al., 2019](#)) and AraBERT ([Antoun et al., 2020](#)). Furthermore, to mitigate the impact of class imbalance, we incorporated class distribution-based weighting. For preprocessing, we compiled Arabic and English punctuation, removed Arabic diacritics via regex ¹, eliminated repeated characters, English words, and numbers, and collapsed multiple spaces into one for clean tokenization. Arabic characters were standardized to reduce variations, ensuring a consistent representation of letters that look or sound similar; for example, different forms of Alif (ا, آ, إ, ؤ) were replaced with the standard form ا (U+0627).

For feature extraction, we used TF-IDF ([Jalilifard et al., 2021](#)) with the top 5,000 terms (unigrams and bigrams). AraBERT and XLM-RoBERTa embeddings were integrated with a 128-token limit, applying padding and truncation, and extracting the [CLS] token from the final hidden state. To fine-tune GPT-5 ([Daniel Han and team, 2023](#)), we employ LoRA adapters within the PEFT

¹<https://docs.python.org/3/howto/regex.html>

framework, incorporating a curated set of few-shot examples.

For Task 3, we employed the CLIP via Radford et al. (2021) processor for feature extraction, utilizing the ViT-B/32² transformer architecture as the image encoder and a masked self-attention transformer as the text encoder. The extracted multimodal features were fed into a Support Vector Machine for classification; it failed to identify hateful memes accurately. The main challenge was dataset (Alam et al., 2024a), (Alam et al., 2024b) imbalance, which could be mitigated by collecting more hateful memes for a balanced distribution. Additionally, as non-Arabic speakers, understanding the language and cultural context was difficult, so we relied on a CNN-based neural network for better performance.

To achieve our objective of developing a lightweight model, we employed the ELUNet architecture via Deng et al. (2022). Since all captions in the dataset are in the Arabic language, textual features were extracted using the DistilBERT tokenizer via Devlin et al. (2018). In the case of preprocessing and cleaning, the same procedure as Task 2 was followed. Another challenge we faced was that the tokenizers’ lengths were not equal for all memes, as they hold different sizes of text. So we fixed the tokenizer size to 256. If the tokenizer length is smaller than the value, the previous value will repeat; otherwise larger size tokenizer will be shrunk using the PCA algorithm (Drikvandi and Lawal, 2023). The corresponding images were processed through the encoder component of the ELUNet architecture. Inspired by Li et al. (2024), a cross-attention mechanism was then applied, integrating the encoded image features from the encoder with the textual embeddings generated by the tokenizer, positioned at the intermediate layers of ELUNet. The cross-attention outputs were subsequently passed through the decoder component of ELUNet. The proposed model (Figure 1) produces two outputs.

4 Experimental Setup

4.1 Emotion, Offensive Language, and Hate Detection

The whole dataset was split into Train(70%), Test(15%), and Validation(15%) via stratified sampling across emotion, hate, and offensive tasks,

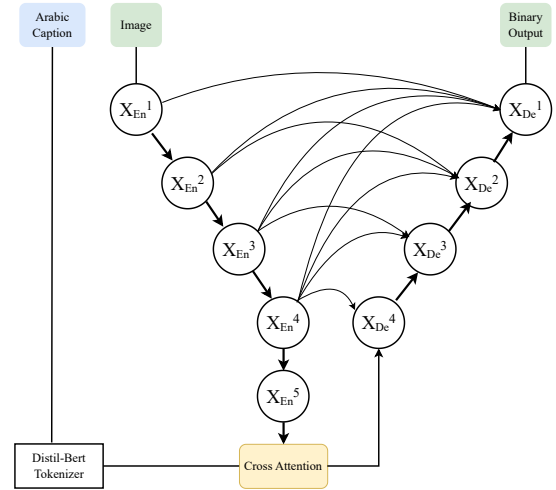


Figure 1: The architecture of Attention-based ELUNet

with exception for GPT-5 (80-10-10). Table 1 provides a brief overview of various emotions in the dataset, including its size and distribution of various emotions, as well as there are offensive (yes - 1744, no - 4216) and hate (yes - 303, no - 1441). Table 2 reveals the Task 2 dataset contains the most non-Arabic characters (see Figure 2).

Name	Amount
Anger	1551
Disgust	777
Neutral	661
Love	593
Joy	533
Anticipation	491
Optimism	419
Sadness	335
Confidence	210
Pessimism	194
Surprise	143
Fear	53

Table 1: Emotion Proportions in Training Data – Task 2

We used Optuna with a class-weighted objective to optimize XGBoost hyperparameters for the highest macro F1-score. We incorporated a deep learning approach using AraBERTv2³ for multitask classification across emotion, offensive language, and hate speech tasks. Three task-specific linear layers mapped the 768-dimensional hidden representation to class logits, with dropout applied to improve generalization. For fine-tuning GPT-5, we

²<https://huggingface.co/openai/clip-vit-base-patch32>

³<https://huggingface.co/aubmindlab/bert-base-arabertv2>

Name	Non-Arabic Chars Count
Train (Task 2)	157138
Test	32968
Validation	32075
Train (Task 3)	4737
Test	1340
Validation	1310

Table 2: Non-Arabic Characters in Tasks 2 & 3

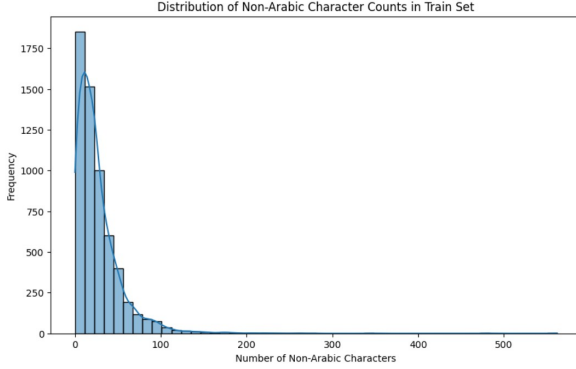


Figure 2: Non-Arabic Character Distribution – Train Set (Task 2)

configured the rank, selected specific transformer layers, and applied an appropriate scaling factor, while enabling gradient checkpointing to optimize memory usage. Furthermore, no bias parameters were introduced to ensure that the fine-tuning process remained lightweight.

	Emo	Offn	Hate
learning rate	0.0060	0.0011	0.0037
max depth	10	7	10
num. estimator	50	282	182
subsample	0.9453	0.8231	0.7524
colsample_bytree	0.7366	0.6489	0.8440
scale_pos_weight	x	2.4179	0.0535

Table 3: Best parameter value from trial run

4.2 Multimodal Hate Speech Detection in Memes

Table 4 presents the distribution of hateful content in training, development, and test sets. We processed each meme (text + image) using CLIP to create joint features. Text was tokenized and images scaled to RGB via CLIPProcessor, producing tensors for both modalities. Features were concatenated and fed to a 1D-CNN. Then we evaluated our enhanced ELUNet model with AraBert, DistilBert tokenizers. Our best model, ELUNet with the

DistilBert tokenizer gave the accuracy of 75%. In our experiment, we chose batch size 16, epoch 5, and learning rate 10^{-3} . This model was trained in Google Colab and consumed 6.2 GB of GPU.

Name	Hate	Not Hate
Train	213	1930
Dev	31	281
Test	154	452
Total	398	2663

Table 4: Dataset Size – Task 3 (Initial)

5 Results

Table 5 summarizes our model’s performance on the task 2 dataset. The results indicate that applying class weights improves performance based on the average F1 score, while incorporating deep learning approaches yields even higher results. For instance, in our experiments with AraBERT, using a batch size of 8, 5 epochs, a dropout rate of 0.3, and a learning rate of 10^{-5} with the exception (10^{-4}) for Gpt-5, we achieved an F1 score of 0.67. Reducing dropout to 0.1, while doubling both batch size and epochs, increased the score to 0.68, matching the performance of DistilBERT. However, with respect to accuracy, GPT-5 and AraBERT achieved comparable performance on the offensive and hate detection tasks, while exhibiting notable differences in the emotion classification task.

App.	Model	Emo	Offn	Hate	Avg
Without Weight	XGB	0.172	0.416	0.344	0.312
	XGB-AraBERT	0.241	0.712	0.541	0.484
	XGB-AraBERT+XLMRoBERTa	0.244	0.414	0.500	0.384
	SVM(Baseline)	0.284	0.702	0.564	0.513
With Weight	XGB	0.212	0.712	0.400	0.393
	XGB-AraBERT+XLMRoBERTa	0.264	0.723	0.500	0.493
	XGB-AraBERT+XLMRoBERTa Trial	0.324	0.775	0.624	0.574
DL	AraBERT	0.267	0.834	0.954	0.684
	DistilBERT	0.373	0.774	0.924	0.683
	Gpt-oss-20b (PC)	0.014	0.412	0.483	0.300

Table 5: Performance of the models on the Task 2 dataset. Here, PC, Emo, Offn, Hate, and Avg denote the post-competition, emotion, offensive, hate, and average macro F1 scores, respectively.

The model performances in Task 3 are described in Table 6. For adding class weight, the result has been improved. Finally, we get an accuracy of 75%. For each testing section test dataset was utilized. Despite fixing the epoch to 20, the best-fitting model took only 5 epochs by using the early stopping concept.

Model	Acc	MacroAvg-f1	Hateful(f1)	Non-Hateful(f1)
ID-CNN(Baseline)	0.745	0.431	0	0.851
ELUNet-DistilBert	0.746	0.421	0	0.852
ELUNet-AraBert	0.744	0.422	0	0.853
ELUNet-AraBert (WW)	0.746	0.372	0	0.855
ELUNet-DistilBert(WW)	0.754	0.500	0.165	0.858

Table 6: Performance of the models on the Task 3 dataset. Here, WW represents 'with weight'.

6 limitations

Both subtasks (Task 2: Emotion, Offensive, and Hate Detection; Task 3: Multimodal Hateful Meme Detection) suffered from severe class imbalance. This led to biased models, poor performance on minority classes, and necessitated mitigations such as class weighting, which still did not fully resolve the issue. Fine-tuning was limited (e.g., 5 epochs with early stopping, a fixed tokenizer length of 256, and PCA for shrinkage), which may have led to underfitting. GPT-5 experiments were constrained by few-shot examples and memory optimizations (e.g., LoRA adapters), resulting in lower emotion detection scores (F1=0.014).

7 Conclusion

Our participation in MAHED 2025 highlights the effectiveness of advanced NLP and multimodal methods for detecting hate speech, hope speech, and emotions in Arabic. For Task 2, our fine-tuned AraBERT scored 0.68 macro F1, surpassing SVM and XGBoost baselines through class-weighted optimization and fused embeddings to address imbalance and noise. For Task 3, our lightweight ELU-Net, cross-attention with tokenizer generated from DistilBert, achieved 75 % accuracy on hateful meme classification despite imbalance. Challenges included limited Arabic meme data, non-Arabic characters, and noisy text affecting preprocessing and features. Future work will explore data augmentation, advanced multimodal fusion, and improved preprocessing and fine-tuning to boost robustness and generalization.

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